

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B. Tech. (CL) Examination
May 2014

CL 309 Construction Planning and Management

Date: 13.05.2014(Tuesday)

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What is role of Project Manager in a construction Firm? 04
(b) Explain the elements of owning cost of equipment. 05
- Q - 2 (a) How the probability of completion time for a project can be determine? Discuss steps involve for it. 04
(b) Compute early start, early finish, latest start and latest finish time for following network in tabular format. 10

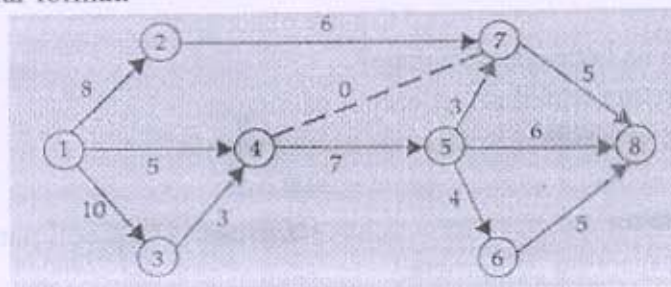


Figure 1.

OR

- Q - 2 (a) Construct an activity network for a project for following details. 08
i. Indicate Critical path.
ii. Calculate t_e , t_l , t_o and free float. Time given is in weeks.

Activity	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Time	4	1	1	1	6	5	4	8	1	2	5	7

- (b) Define: 06
Direct cost, Indirect cost, Critical path, Crash cost, Cost-slope, Activity
- Q - 3 (a) Determine work breakdown for the casting concrete beam over an opening. 04
(b) What do you understand by a 'dummy'? What is its use? 04
(c) Explain Fulkerson's rule of numbering the events in a network. 04

OR

- Q - 3 (a) What do you understand by work breakdown structure? What is its importance in network planning? 04

- (b) Differentiate between 'forward planning' 'backward planning'. 04
- (c) Define the following : 04
- Event, Head event, Tail event, Dual role event.

SECTION - II

- Q - 4 (a) What is importance of organization for construction projects? State merits and demerits of line and staff organization. 04
- (b) What is project Monitoring and Controlling? What do we monitor in project? 05
- (c) Explain the system of wages for Indian labours. 02

- Q - 5 (a) Explain phases of management. 06
- (b) State causes of failures of construction projects. 06

OR

- Q - 5 (a) Explain Objectives of management, 06
- (b) List and explain management methods and techniques 06

- Q - 6 (a) Explain the effect of Slope, Altitude and temperature on performance of I.C. engine. 06
- (b) Regarding scraper, answer the following questions. 06

- i. Draw neat figure of scraper.
- ii. Where one can recommend the use of scraper?
- iii. What is its main disadvantage?
- iv. State factors which affect output of scraper.
- v. Suggest techniques to increase output with it.

OR

- Q - 6 (a) Write short note on compacting equipments. Discuss how soil parameters govern the selection of compacting equipment. 06
- (b) State types and name of hauling equipments. What is the importance of hauling equipment in construction Industry? 06
